SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 3.3: Introduction to Electronics — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.3: Introduction to Electronics
Driving Question	DRIVING QUESTION: Why does a silicon solar panel on a Kajiado farm produce less power in the hot afternoon sun than on a cool morning — and what is so special about silicon that it can act as both a conductor and an insulator depending on the conditions?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **The Solar Panel Puzzle — Encountering the Phenomenon & Sorting Materials**	Students examined voltage and current readings from a Kajiado farm solar panel recorded at 7 a.m. (cool, ~18 °C) and 2 p.m. (hot, ~38 °C). Data consistently showed higher power output in the morning despite brighter afternoon sunlight. Students also sorted household and lab materials (copper wire, rubber, graphite pencil rod, distilled water, salt water, silicon wafer) into 'conductor', 'insulator', or 'unsure' using a simple circuit tester, discovering that silicon and other materials did not fit neatly into either category.	Materials that conduct electricity partially — with electrical conductivity between that of conductors and insulators — are called SEMICONDUCTORS. Silicon is a semiconductor. At room temperature it conducts some current; at very low temperatures it behaves almost like an insulator. This dual behaviour is the first clue to the Kajiado solar panel puzzle. The conductivity of a semiconductor increases with temperature (unlike metals, which become more resistive when hot). Key vocabulary established: conductor, insulator, semiconductor, resistivity, silicon (Si), germanium (Ge).	Pedagogical note — this lesson is purely phenomenon-anchoring; resist the urge to over-explain mechanisms. The goal is productive confusion. Cold-call at least 4 students to articulate what surprised them. Post their unanswered questions directly onto the Driving Question Board (DQB). Expected misconception to surface and park (do not resolve yet): 'More sunlight = more power always.' Validate the observation, then ask, 'What else is changing between morning and afternoon besides light intensity?' Wait 30 seconds before taking responses.
Lesson 2 **Atomic Structure of Silicon — Covalent Bonding and the Crystal Lattice**	Students built physical or digital models of silicon atoms (atomic number 14, four valence electrons) and arranged them into a tetrahedral covalent lattice, mimicking the diamond-cubic crystal structure. They observed that every silicon atom shares all four valence electrons with four neighbours, leaving no electrons obviously 'free' to carry current — yet the Lesson 1 circuit tester showed silicon does conduct a little.	Silicon's four valence electrons form four covalent bonds with neighbouring silicon atoms in a rigid, three-dimensional crystal lattice. At absolute zero (0 K) all electrons are locked in bonds — silicon would be a perfect insulator. At room temperature, random thermal vibrations occasionally break a covalent bond, freeing an electron and leaving behind a positively charged 'hole'. Both the free electron and the hole	Pedagogical note — use the analogy of githeri on a jiko: at low heat (low temperature) the maize kernels (electrons) stay tightly packed; as heat increases, kernels 'pop' free (thermal generation of electron-hole pairs). Have students draw the covalent lattice with at least 5 silicon atoms and mark one broken bond with a free electron (e^-) and a hole (h^+). Connect back to DQB: 'We now know WHY silicon

	Students then compared the silicon lattice model to a copper lattice (one free electron per atom) and a rubber polymer model (electrons tightly localised).	can move through the lattice and carry charge. The higher the temperature, the more bond-breaking occurs, so conductivity rises with temperature. This explains the semiconductor identity observed in Lesson 1.	conducts a little — but does this explain the afternoon power drop yet?' (Answer: partially — more heat → more free carriers, so conductivity should improve, yet panel power drops. Flag this tension explicitly.)
Lesson 3 **Intrinsic Semiconductors — The Energy Band Model and Silicon's Dual Identity**	Students examined energy band diagrams for a conductor (copper), insulator (rubber/SiO_2), and semiconductor (silicon), comparing band-gap sizes. A simulation (or teacher-drawn diagram) showed electrons in silicon jumping the ~1.1 eV band gap when thermal energy is supplied, landing in the conduction band and leaving holes in the valence band. Students plotted a sketch graph of conductivity vs. temperature for silicon vs. copper, noting the opposite trends.	Solids can be classified by their energy band structure: conductors have overlapping or partially filled conduction and valence bands (band gap ≈ 0 eV); insulators have a large band gap (> 5 eV) that electrons cannot cross at ordinary temperatures; semiconductors have a small but non-zero band gap (silicon: ~1.1 eV, germanium: ~0.67 eV) that electrons can cross with moderate thermal or photon energy. When an electron crosses the gap it becomes a free carrier in the conduction band; the vacancy it leaves in the valence band is the hole. In an intrinsic (pure) semiconductor, free electrons and holes are always generated in equal numbers. Increasing temperature increases the number of electron-hole pairs — increasing conductivity. This is the OPPOSITE trend to metals.	Pedagogical note — the band gap concept is the most abstract in this sub-strand. Anchor it concretely: 1.1 eV is approximately the energy of a near-infrared photon — this is why silicon absorbs sunlight and is useful in solar panels. Draw the band diagram on the board and require every student to copy and label it (formative check). Cold-call 3 students to predict: 'If silicon's band gap were 5 eV like glass, could it be used in a solar panel?' Expected answer: No — visible light photons (~1.8–3.1 eV) would still be enough, but the point is that a very large gap means most photons cannot excite electrons. Also revisit the copper-vs-silicon conductivity graph: metal conductivity FALLS with temperature (more lattice vibrations scatter electrons); semiconductor conductivity RISES. Post this distinction on the DQB as a partial answer to the driving question.
Lesson 4 **Doping Silicon: How Impurities Turn a Crystal into a Switch**	Students examined diagrams showing a phosphorus atom (5 valence electrons) substituted into the silicon lattice. The fifth electron has no bond partner and is only weakly held — it becomes a free electron at room temperature without needing to break a covalent bond. Students then examined a boron atom (3 valence electrons) substituted into the lattice: the missing fourth bond creates a hole at room temperature. Comparing resistivity values of pure silicon (~640 Ω·m) vs. phosphorus-doped silicon (~0.01 Ω·m) made the dramatic effect of doping vivid.	Doping is the deliberate addition of a tiny quantity of impurity atoms to a pure (intrinsic) silicon crystal to control its conductivity. (1) N-TYPE semiconductor: doping with a Group V element such as phosphorus (P) or arsenic (As) donates one extra free electron per dopant atom. The dopant is called a DONOR. Majority carriers are electrons; minority carriers are holes. (2) P-TYPE semiconductor: doping with a Group III element such as boron (B) or aluminium (Al) creates one hole per dopant atom. The dopant is called an ACCEPTOR. Majority carriers are holes; minority carriers are electrons. Doping concentrations of even 1 part per million can increase conductivity by factors of thousands. The silicon in Kajiado solar panels is made of layers of n-type and p-type doped silicon.	Pedagogical note — a powerful Kenya-context analogy: doping silicon with phosphorus is like adding a pinch of salt (chumvi) to ugali water — a tiny addition dramatically changes the electrical 'flavour' (conductivity) of the whole batch. Stress that doped silicon is still electrically neutral overall; doping changes the TYPE and NUMBER of charge carriers, not the net charge. Common misconception to address: 'N-type silicon is negatively charged.' Explicitly correct this. Have students draw both n-type and p-type lattice diagrams labelling donor/acceptor atoms, majority carriers, and minority carriers. Connect to DQB: 'Doping creates the two layers of the solar panel — but how does the panel actually push current out?' (Sets up Lesson 5.)

Lesson 5 **The P-N Junction — How a Boundary Between Two Silicon Types Becomes a One-Way Gate**	Students used a diagram or simulation to observe what happens when p-type and n-type silicon are joined: electrons diffuse from n-type into p-type and holes diffuse from p-type into n-type near the boundary, creating a depleted region (the DEPLETION LAYER) that has a built-in electric field pointing from n to p. Students then observed forward bias (external voltage assists current across junction) and reverse bias (external voltage widens depletion layer, blocking current), noting the p-n junction acts as a DIODE — current flows easily in one direction only.	When n-type and p-type silicon are brought into contact, diffusion of majority carriers near the junction creates a depletion layer devoid of free carriers, with a built-in potential difference (typically ~0.6–0.7 V for silicon). This built-in field opposes further diffusion until equilibrium. Under forward bias (positive terminal to p-side), the depletion layer narrows and current flows readily. Under reverse bias (positive terminal to n-side), the depletion layer widens and almost no current flows — the junction rectifies AC to DC. In a solar cell, incoming photons with energy ≥ 1.1 eV generate electron-hole pairs near the junction; the built-in field sweeps electrons to the n-side and holes to the p-side, producing a photocurrent. This is the PHOTOVOLTAIC EFFECT — the operating principle of the Kajiado solar panel.	Pedagogical note — this is the mechanistic heart of the driving question. Spend extra time here. Use the analogy of the Rift Valley escarpment as a built-in 'hill': electrons roll downhill (n→p direction of field) when photons knock them free. Have students trace the path of one photon-generated electron-hole pair from creation → separation by the junction field → exit through the external circuit → recombination. This completes the circuit story. Now the DQB partial answer reads: 'The p-n junction separates photon-generated carriers using its built-in field, producing a voltage.' The afternoon power-drop tension still needs temperature resolution — deliberately leave it for Lessons 6–7.
Lesson 6 **Case Studies — Solar Panels, Microchips & Superconducting Magnets: Applying Semiconductor Understanding to Kenyan Contexts**	Students examined three case studies: (1) Kajiado solar farm data showing panel efficiency vs. panel temperature, confirming ~0.4–0.5 % efficiency loss per °C rise above 25 °C. (2) A simplified cross-section of a MOSFET transistor (the building block of microchips in M-Pesa handsets) showing how a gate voltage controls a conducting channel between doped regions. (3) Data on superconducting magnets used in MRI machines at Kenyatta National Hospital, where resistance drops to exactly zero below a critical temperature — the extreme opposite of the semiconductor temperature trend.	The same family of semiconductor properties — band gaps, doping, p-n junctions, and temperature-dependent carrier behaviour — underpins three transformative technologies: (1) SOLAR CELLS: photovoltaic effect at the p-n junction converts sunlight to electricity; efficiency decreases at high temperatures because increased thermal carrier generation raises recombination rates and reduces the open-circuit voltage (V_OC). This is why the Kajiado panel loses power in the hot afternoon — the junction voltage shrinks. (2) TRANSISTORS (microchips): doped regions controlled by electric fields switch and amplify signals; billions fit on one chip. (3) SUPERCONDUCTORS: certain materials below their critical temperature have zero resistivity — band theory breaks down and a quantum phenomenon (Cooper pairs) takes over. KNH's MRI magnet uses liquid-helium-cooled niobium-titanium wire.	Pedagogical note — this lesson builds scientific literacy and 21st-century skills (connection to real Kenyan infrastructure). Emphasise that the Kajiado power-drop is NOT because the panel 'gets tired' or because clouds appear — it is a fundamental semiconductor physics effect. The key equation for teachers to reference: P_max decreases because V_OC ≈ (kT/q) × ln(I_L/I_0) — as T rises, the logarithmic term changes, and increased dark current I_0 (from thermally generated carriers) reduces V_OC. Students do NOT need the equation; they need the qualitative chain: higher T → more thermal e-h pairs in bulk → higher recombination rate → lower junction voltage → lower power output. Assign the M-Pesa transistor case as a group discussion: 'How is the transistor in your phone related to the silicon in the Kajiado panel?' Cold-call 3 groups for one-sentence answers.
Lesson 7 **Final Explanation: Semiconductors,**	Students revisited all evidence accumulated across Lessons 1–6: the original Kajiado morning-vs-afternoon power data, the	Silicon is a semiconductor because its band gap (~1.1 eV) is small enough for electrons to jump from the valence band to the	Pedagogical note — the final lesson must explicitly close every thread opened in Lesson 1. Use a structured 'claim–

Doping, and the Kajiado Solar Panel Mystery — Resolved	conductivity sorting results, the band diagrams, doping models, p-n junction diagrams, and the case study efficiency-vs-temperature graph. They also reviewed their own DQB entries to confirm which questions had been answered and which (if any) remain open. As a culminating task, each student or group constructed a final annotated model tracing the journey of sunlight → electron-hole pair generation → junction separation → external current → power output, with arrows and labels explaining where and why temperature reduces output.	conduction band when excited by thermal energy or photons, yet large enough that it is not a conductor at room temperature. Doping silicon with phosphorus (n-type) or boron (p-type) controls its conductivity by orders of magnitude. A p-n junction between n-type and p-type silicon creates a built-in electric field that separates photon-generated electron-hole pairs, producing a photovoltaic current — this is how the Kajiado solar panel generates electricity. In the hot afternoon (~38 °C vs. 18 °C morning), increased thermal energy generates extra electron-hole pairs in the bulk silicon that recombine before reaching the junction, raises the dark saturation current, and reduces the junction's built-in voltage (V_{OC}). The net effect is lower maximum power output even under bright sunlight. The Kajiado puzzle is fully resolved: more heat does not help — it undermines the very voltage that drives the current out.	evidence–reasoning' (CER) framework: each student writes one claim ('The afternoon power drop happens because…'), cites at least two pieces of evidence from the lessons, and gives one reasoning sentence linking evidence to claim. Collect as a formative assessment. Also prompt metacognitive reflection: 'Which part of this explanation surprised you most?' and 'What new questions do you now have about electronics?' These new questions can seed Sub-Strand 3.4. Key teacher move: validate that the driving question has TWO parts — (1) why less power in the afternoon (resolved by temperature effect on V_{OC}) and (2) what makes silicon special (resolved by band gap + doping + p-n junction). Both must appear in the final model. Award marks for models that show the morning vs. afternoon comparison explicitly.